Prabhavit Gupta

Software & AI Architect: Agent Orchestration, RAG Systems, Cloud Backends & Reliability

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Summary

Software & AI Architect with 3 years of commercial production experience designing and deploying scalable software architectures, multi-agent AI orchestration platforms, and high-performance RAG pipelines. Hands-on technical background in Python, TypeScript, Node.js, and PostgreSQL, with deep expertise in model context protocols (MCP), tool-augmented agents, and continuous evaluation harnesses. Proven track record leading architectural decisions, integrating AI workloads with enterprise database backends, and ensuring high system availability, security, and maintainability. Based in Venice, Italy with permanent EU work authorization (Permesso di Soggiorno UE a Tempo Indeterminato, immediate start, full remote ready).

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production multi-agent systems and custom LLM harnesses serving 4,500+ active enterprise users with 99.9% uptime.
- Designed modular agentic orchestration frameworks incorporating dynamic tool routing, schema-constrained outputs, and deterministic retry mechanisms.
- Built low-latency Retrieval-Augmented Generation (RAG) pipelines over multi-tenant relational data stores, implementing semantic chunking and hybrid search.
- Structured and optimized relational PostgreSQL schemas, complex indexing, and high-throughput transactional backends using Python and TypeScript.
- Established automated evaluation (evals) and observability suites to monitor model hallucinations, token economics, latency drift, and execution reliability.
- Built robust CI/CD deployment pipelines using Docker containerization and GitHub Actions, ensuring consistent test coverage and zero-downtime releases.
- *Tools: Python, TypeScript, Node.js, Next.js, PostgreSQL, Docker, GitHub Actions, AWS, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Designed automated data ingestion architectures and entity extraction routines connecting external partner APIs with central PostgreSQL repositories.
- Implemented rigorous schema validation layers and data normalization rules to guarantee consistency across multi-source client feeds.
- Collaborated with business stakeholders, technical teams, and external partners to establish architectural blueprints, SLAs, and milestone deliveries.
- *Tools: Python, TypeScript, REST APIs, PostgreSQL, Supabase, Git*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Delivered custom AI-driven automation systems, webhook integrations, and RESTful microservices for Italian and European enterprise clients.
- Engineered agent evaluation harnesses and automated integration tests (pytest) to validate multi-step decision logic prior to deployment.
- Authored technical architecture runbooks, system specifications, and API documentation for distributed engineering teams.
- *Tools: Python, TypeScript, REST APIs, Supabase, Docker, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated external REST API services and email notification workers, establishing automated validation checks across core onboarding flows.
- *Tools: Python, REST APIs, Sendgrid*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization in Italy and the EU (CCNL or Partita IVA). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working beginner).
- **Availability:** Available immediately for full-time remote roles in Italy.

Core Skills

- **AI Architecture & Systems:** Multi-agent orchestration, custom execution harnesses, RAG systems, tool calling (MCP / function calling), continuous validation pipelines.
- **Backend Engineering:** Python 3, Node.js, TypeScript, RESTful API design, microservices architecture, asynchronous event handling, clean code.
- **Data Infrastructure:** PostgreSQL, relational schema design, SQL query optimization, transaction management, data validation (Pydantic / Zod).
- **DevOps & Operations:** Docker containerization, AWS cloud environments, Linux system administration, GitHub Actions CI/CD automation, Git.
- **Leadership & Delivery:** Technical architecture design, requirements definition, cross-functional collaboration, technical documentation, agile delivery.